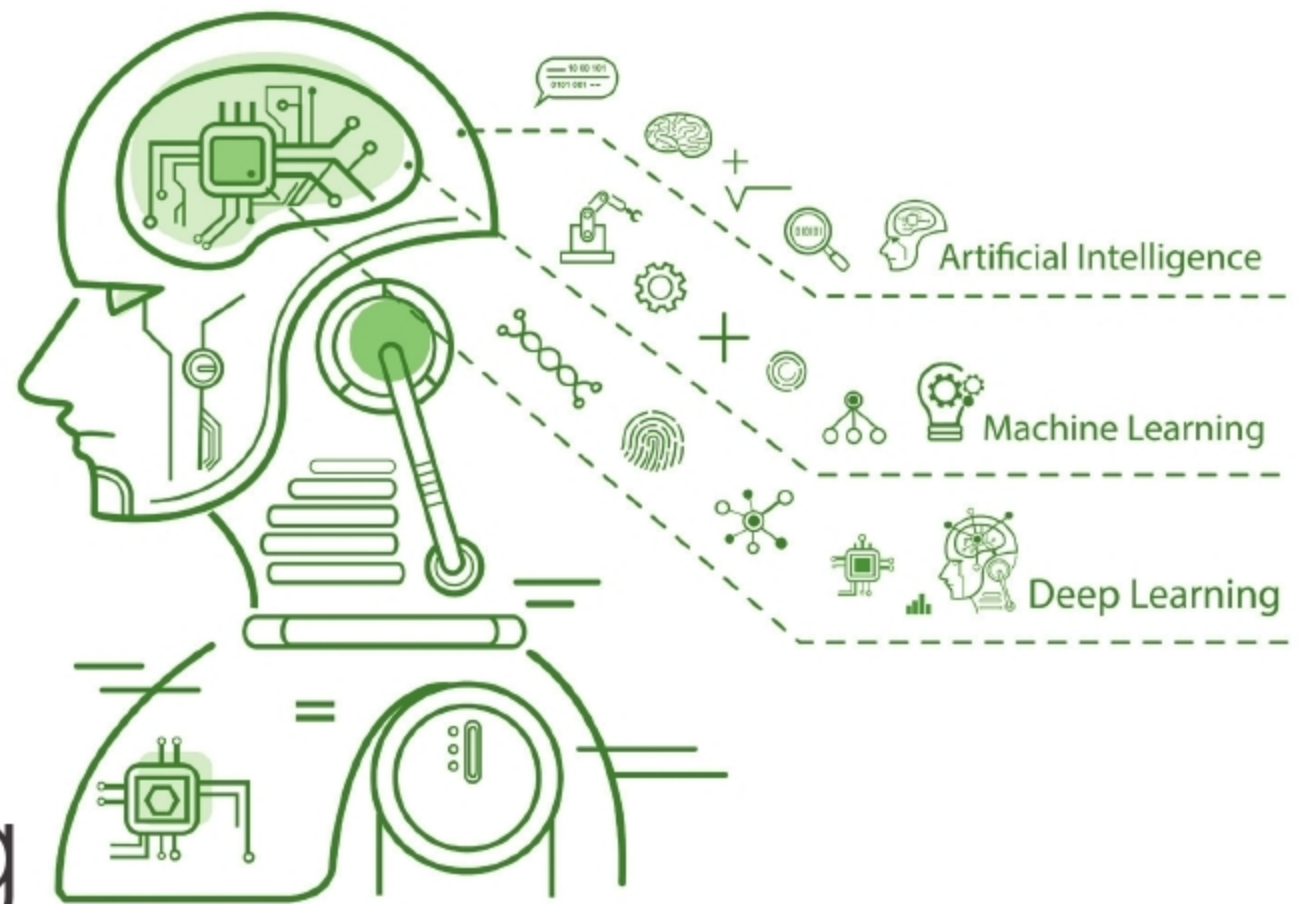




Applying Machine Learning to IoT

Machine learning in conjunction with IoT will play an increasingly important role in our lives as the days go by, as both are fields of computer science that are currently in a rapid state of development.

We are undoubtedly becoming 'smarter' in various aspects of our daily life, with IoT and machine learning (ML) playing a crucial role in this. Well known thought leaders like Bill Gates and Dr Judith Dayhoff (author of 'Neural Network Architectures: An Introduction') say that the IoT has given our physical inanimate world a digital nervous system. IoT has really exploded over the past three years, demonstrating its potential in applications ranging from wearables and automated cars to smart homes and smart cities, creating an impact everywhere. According to recent research by Gartner, there are around 16 billion devices connected to the IoT now and this is expected to rise to 25 billion by 2020. All such connected devices generate a deluge of information that needs to be monitored and analysed, so that they learn continuously from the available sets of data and improve themselves without any manual intervention. That's how IoT devices are becoming smarter.

So how will such a large ocean of data get analysed and monitored? This is where the role of machine learning comes in. There are different ML algorithms and techniques that are implemented to easily analyse massive amounts of data in a short span of time, increasing the efficiency of the IoT. Also, different ML techniques such as decision trees, clustering, neural and Bayesian networks, help the devices to identify patterns in different types of data sets coming from diverse sources, and take appropriate decisions on the basis of their analysis. Such challenges are faced especially in the case of embedded systems. The most important thing is that there is no programming or coding support given to these devices all through this process. Hence, it would not be wrong to say that if the IoT is the digital nervous system, then ML acts as its medulla oblongata. Without implementing ML, it would really be difficult for smart devices and the IoT to make smart decisions in real-time, severely limiting their capabilities.